1. Clinical Question & Core Hypothesis

Clinical Question

Although ERBB2-amplified (HER2-positive) tumors are treated with trastuzumab, not all immune-infiltrated tumors (high TILs) derive clinical benefit; what tumor-intrinsic mechanisms disable the functional advantage of immune infiltration? 2) Tier 1 paths used: Core Path 1 (trastuzumab response tied to specific molecular biomarkers) motivates focusing on therapy response; Core Path 2 (IHC4+C/pCR) introduces proliferation and IHC context (supports inclusion of MKI67 as a modifier); Core Path 3 (PTEN expression) provides a mechanistic axis—loss of PTEN → PI3K/AKT activation—that plausibly causes immune resistance and reduced trastuzumab efficacy. I built the hypothesis on the intersection of these paths because they link target dependence (ERBB2), tumor-intrinsic signaling (PTEN/PI3K), and measurable immune context (TILs) to therapy response. 3) Tier 2 evidence that supports the claim: BIRC5 in CTCs suggests apoptosis-resistant disseminated clones can persist despite systemic HER2 blockade; ACSL4 indicates metabolic reprogramming that can modulate immune interactions; IHC4 components (including MKI67) and radiomic features provide orthogonal readouts to stratify these phenotypes. These support testing whether PTEN loss functionally neutralizes TIL-associated benefit and whether additional markers (BIRC5, ACSL4, Ki-67) refine that effect. 4) Endpoint choice rationale: I selected therapy_response (HER2-targeted therapy response) as the single primary endpoint because the core clinical problem is failure to benefit from trastuzumab; therapy response is actionable, directly tied to the core paths (trastuzumab→pCR/response), and measurable in neoadjuvant/clinical contexts (pCR, imaging/clinical response). 5) Deliberate exclusions and why: I excluded RS (recurrence score) and NTX as they are largely prognostic or metastatic-burden indicators rather than mechanistic mediators of trastuzumab-specific immune resistance; radiomic biomarkers and CTC-based BIRC5 remain secondary modifiers rather than core variables to retain sharp mechanistic focus. I also excluded weaker semantic chains (troponin→ACSL4 via ELISA) from the core hypothesis and relegated ACSL4 to secondary_modifiers for exploratory testing?

Working Hypothesis

In ERBB2-amplified (HER2+) breast cancers with high intratumoral TIL density, PTEN loss creates a tumor-intrinsic resistance to trastuzumab by activating PI3K/AKT signaling and promoting immune evasion, such that high TILs do not translate into robust therapy response. ERBB2 (HER2) and PTEN serve as anchor genes; Tier 2 modifiers BIRC5 (survivin), ACSL4, and MKI67 modulate the magnitude of this resistance by enhancing anti-apoptotic, metabolic reprogramming, and proliferative traits, respectively. Clinically, tumors that are TIL-high and PTEN-null will show reduced response to HER2-targeted therapy, and the magnitude of resistance will be greater in samples with high BIRC5 and/or ACSL4 expression and/or high MKI67. This framework predicts that adding PI3K pathway inhibitors or immune-modulatory strategies to trastuzumab could restore or improve treatment response in this distinct molecular/immune context.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	ERBB2, PTEN
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERBB2: ERBB2-amplified (HER2-positive) tumors are treated with trastuzumab, not all immune-infiltrated tumors (high TILs) derive clinical benefit; what tumor-intrinsic mechanisms disable the functional advantage of immune infiltration? 2) Tier 1 paths used: Core Path 1 (trastuzumab response tied to specific molecular biomarkers) motivates focusing on therapy response; Core Path 2 (IHC4+C/pCR) introduces proliferation and IHC context (supports inclusion of MKI67 as a modifier); Core Path 3 (PTEN expression) provides a mechanistic axis—loss of PTEN → PI3K/AKT activation—that plausibly causes immune resistance and reduced trastuzumab efficacy.

PTEN: PTEN expression) provides a mechanistic axis—loss of PTEN → PI3K/AKT activation—that plausibly causes immune resistance and reduced trastuzumab efficacy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	BIRC5	Supporting modifier from knowledge graph
Secondary	ACSL4	Supporting modifier from knowledge graph
Secondary	MKI67	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective clinical analysis of HER2+ tumors with annotated TIL density, PTEN status (IHC/sequencing), BIRC5/ACSL4/MKI67 expression, and trastuzumab response outcomes (pCR, ORR, PFS) to test associations and interactions.

Prediction 2: In vitro assays using HER2+ cell lines with CRISPR/Cas9-mediated PTEN knockout or knockdown to model PTEN loss; assess trastuzumab-mediated ADCC in the presence of NK cells, with and without PI3K inhibitors (e.g., alpelisib) and measure apoptosis (Annexin V/PI), PI3K/AKT signaling (pAKT), and immune synapse activity.

Prediction 3: In vivo models (xenografts or patient-derived xenografts) with defined PTEN status to evaluate tumor growth, ADCC markers, immune infiltration, and response to trastuzumab ± PI3K inhibitors.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- BIRC5: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ACSL4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.